SignalWard Project Proposal

TEAM RADHE | AI/ML · IoT SIMULATION · SOFTWARE PRODUCT DEVELOPMENT

THE IDEA	An evidence-aware home-care digital twin that separates patient-pattern signals from sensor trust, so noisy or stale readings trigger a measurement check instead of a misleading patient-risk score.

The problem

Home monitoring combines vital readings, caregiver observations, and medical-equipment status, but these signals may be incomplete or misleading. One abnormal value can reflect a patient change, a poor measurement, stale data, or a device/power issue. A dashboard that treats every value as equally trustworthy can create confusing alerts, while connectivity interruptions can delay review.

Our solution

SignalWard simulates a home-care episode and routes each packet through three separate views: patient trajectory, evidence confidence, and equipment context. It explains the resulting event and supports caregiver acknowledgement, notes, and visit logging. A browser queue demonstrates store-and-forward behavior during simulated outages.

Why this approach stands out

- Evidence before risk: low-quality or stale packets suppress the patient-state score and create an Evidence Check.
- A visible counterexample: the Noisy Sensor scenario looks alarming but has poor evidence confidence; compare it with clean multi-signal drift.
- Separate fault domains: device connection and backup battery remain equipment context, not patient physiology.
- Human review stays central: explanations and acknowledgement are visible; the system does not prescribe treatment.

End-to-end architecture

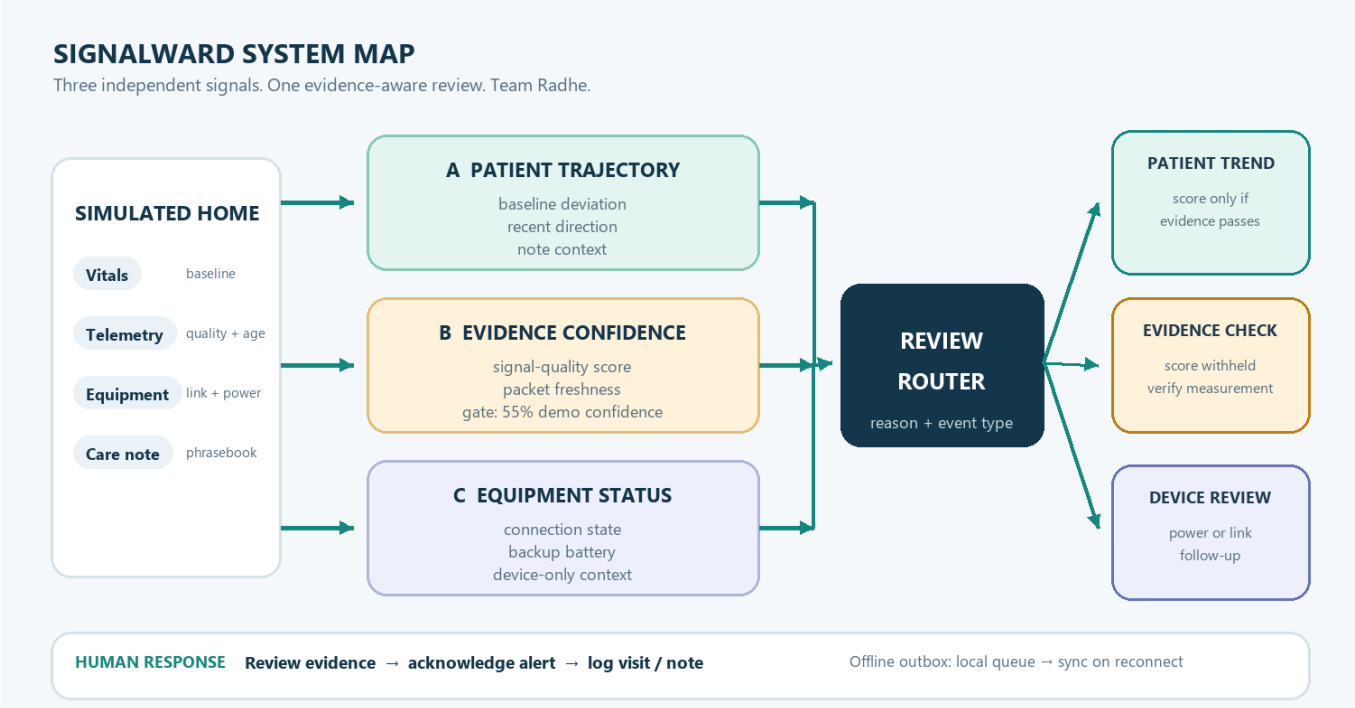


Figure 1. Three signal tracks are routed separately before a caregiver reviews the event.

The key safety decision

A suspicious value is not enough. SignalWard first asks whether the measurement is trustworthy.

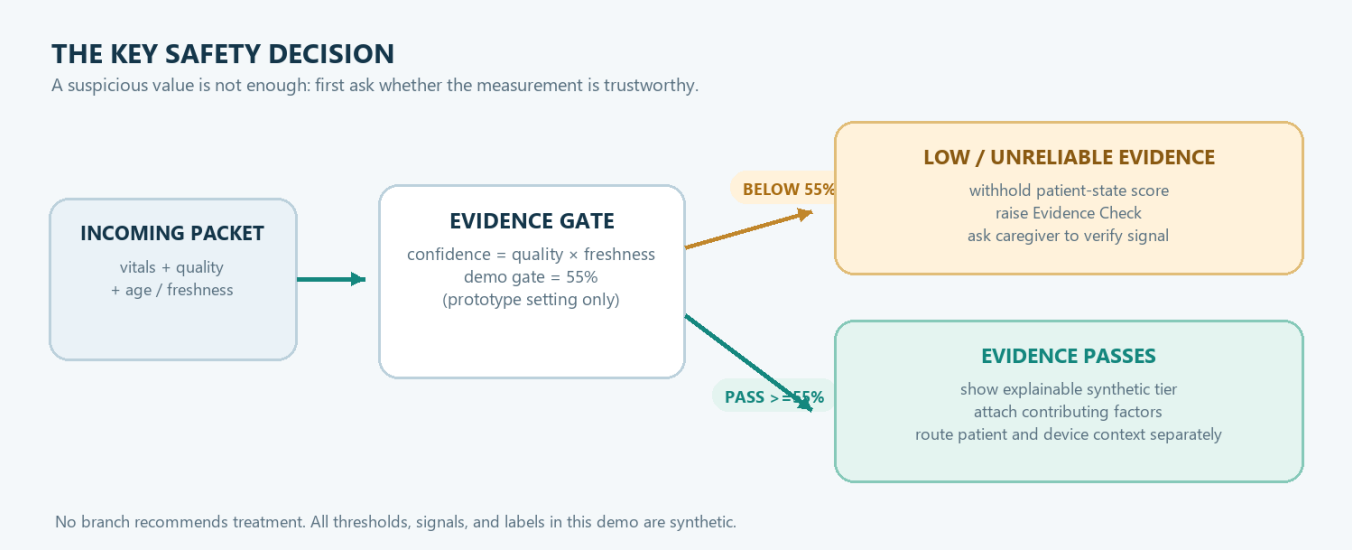


Figure 2. Below the demo confidence gate, the patient-state score is withheld and an Evidence Check is raised.

Demo scenarios

SCENARIO	WHAT IT SHOWS	EXPECTED ROUTE
Stable	Near-baseline readings, reliable mock signal	No elevated synthetic trend
Watch	Clean, moderate multi-signal drift	Illustrative trend + reason labels
Noisy Sensor	Concerning-looking value, poor signal quality	Evidence Check; patient score withheld
Escalation	Larger synthetic drift + device interruption	Combined review with device context separated
Power Backup	Near-baseline readings + backup-power state	Device context visible

Caregiver response and resilience

The local-first workflow keeps events visible and records the caregiver response during simulated connectivity loss.

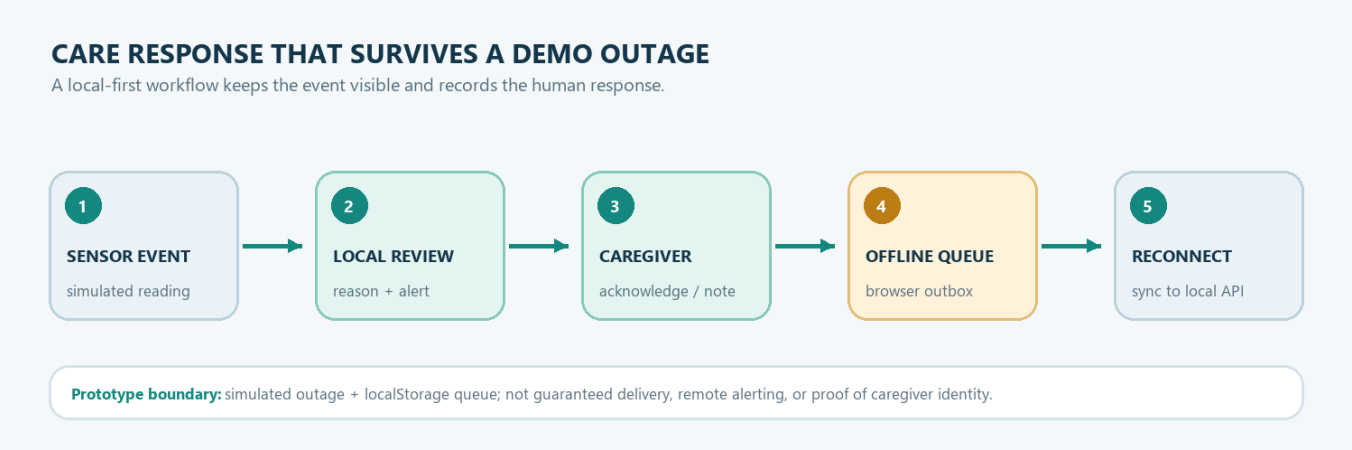


Figure 3. The browser outbox queues demo actions during simulated disconnection and syncs after reconnect.

Technology stack

COMPONENT	IMPLEMENTATION
Frontend	Responsive HTML, CSS, vanilla JavaScript; SVG trend chart
API + simulator	Python standard-library HTTP server; built-in synthetic scenarios
Persistence	SQLite for readings, notes, alerts, acknowledgements, and attendance events
Model demo	Logistic regression trained at startup on generated examples; separate evidence gate
Offline behavior	Browser localStorage outbox; simulated equipment backup-power telemetry
Deployment	Localhost only; no hardware, cloud service, or external API

Expected outcomes and safety

A concise evaluation story for the final presentation.

Expected outcome

Demonstrate the full flow from simulated sensor packet to reasoned event to caregiver response. Judges can inspect these scenario behaviors:

- Noisy sensor routes to Evidence Check and withholds the patient-state score.
- Stable readings remain without an elevated synthetic pattern.
- Clean drift displays a synthetic tier and contributing factors.
- Device interruption remains separate equipment context.
- Caregiver acknowledgement is recorded; offline actions sync after reconnect.

Safety, privacy, and technical limits

All profiles, readings, notes, signal-quality values, equipment states, and model labels are generated. The model is trained only on generated examples and is not clinically validated. Its "24–48 hour" label is a demo target, not a real forecast. A synthetic quality score cannot prove sensor accuracy. The demo caregiver code does not prove identity or physical presence. Do not enter real patient data. The local prototype is not designed for internet exposure and does not send real notifications or make care decisions.

TEAM RADHE	Project: SignalWard — Confidence-Aware Home-Care Digital Twin | Submission: IoT simulation + full-stack platform + synthetic AI/ML demonstration
PRESENTATION LINE	"A bad reading should first trigger a check on the evidence. SignalWard makes that distinction visible, explainable, and testable."